

1 • Shared evidence: comparison bureau

The life-cycle comparison bureau

Name _____ Date _____

Named-species summaries. Use these cards for W1-W3 and your selected route.

F • Common frog

Eggs are laid in pond water. Tadpoles are larvae with tails; they develop into froglets and then adults. No pupa stage. No milk feeding.

P • Platypus

A mammal that lays eggs in a burrow. The young hatch and receive milk from their mother; they grow to reproductive maturity. No pupa.

Comparison rule

Use the same feature on both sides: egg location, pupa stage, or milk feeding. A blank in our evidence means not supplied, not automatically absent.

M • Monarch butterfly

Eggs are laid on milkweed. Caterpillars are larvae; they become pupae, then adult butterflies. No milk feeding.

R • Robin

A bird that lays eggs in a nest. Chicks hatch, grow feathers and become fledglings, then reproductive adults. Parents bring food; no milk feeding.

Cycle rule

Adults produce offspring. The next eggs belong to a new generation; the same individual does not turn back into an egg.

W1. Group the four species by pupa stage, then by milk feeding. Why do groups change?

W2. Compare frog and monarch: one similarity and one difference.

W3. Give a counterexample to “all egg-layers are birds”.

2 • Supported route: compare a pair

Use cards F and M. Word bank: eggs / larva / pupa / adult.

S1. Which stage occurs in both routes?

A. Egg

B. Pupa

S2. Complete: Both have a _____ stage. The monarch has a _____ ; the common frog does not.

S3. Put these monarch stages in order: adult, egg, pupa, caterpillar.

S4. A platypus lays eggs. Is it a bird?

A. Yes

B. No: it is a mammal

S5. Explain why “all egg-layers are birds” is wrong. Name one card.

3 • Standard route: feature table

Use the same question for each animal.

Species	Pupa present?	Young receive milk?
Common frog		
Monarch		
Platypus		
Robin		

N1. Complete the table using the species cards.

N2. Compare frog and monarch using one similarity and one difference.

N3. Compare platypus and robin using egg-laying and feeding of young.

N4. Correct “all egg-layers are birds”. Explain why four species do not describe every animal.

4 • Stretch route: challenge the rules

A strong claim stays within the evidence.

T1. Write a comparison of two species using three matched features.

T2. Evaluate “all larvae become pupae” using the frog and monarch cards.

T3. Evaluate “all mammals give live birth”. Identify a counterexample and its classification evidence.

T4. A fifth species has no feeding information. Can you record “no parental feeding”? Explain a better entry.

5 • Exit and reflection

Answer independently using scientific features.

E1. Give one similarity between frog and monarch cycles.

E2. Give one difference in their stages.

E3. Give a counterexample to “all egg-layers are birds”. Explain.

E4. Does an adult turn into its own egg? Explain the cycle arrow.

Reflection. Which comparison question would you investigate next?
